When Trade Preferences Become Easier to Use: Evidence from Jordan

Ayman Tibi*

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Abstract

Trade agreements often condition preferential market access on non-tariff eligibility requirements, such as rules of origin. These requirements can limit the use of preferential agreements and hinder deeper trade integration. In this paper, I study how trade responds when exporters gain access to a less restrictive preferential trade regime. I exploit the implementation of the U.S.–Jordan Free Trade Agreement (FTA), which introduced an alternative to Jordan's existing Qualifying Industrial Zones (QIZ) program. Both regimes ultimately provided tariff-free access to the U.S. market, but the FTA did so under less restrictive rules of origin and production-location requirements. Using product-level trade data from multiple sources, I find that the FTA's less restrictive eligibility requirements affected trade along three margins. Exporters shifted away from the QIZ program and toward the FTA, exports to the United States increased in industries with a strong pre-existing presence in that market, and FTA liberalization induced new product entry into the U.S. market. The results suggest that reducing non-tariff barriers embedded in preferential trade agreements can deepen trade integration by encouraging the use of less restrictive preference regimes and expanding trade at both the intensive and extensive margins.

Keywords: Rules of origin, Non-Tariff Barriers, Trade Liberalization, Free Trade Agreements, MENA

*University of Mannheim, atibi@mail.uni-mannheim.de

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1 Introduction

Free trade agreements are an important feature of global trade and production (Baier and Bergstrand, 2007; Mattoo et al., 2022). A large literature has examined how trade liberalization through free trade agreements and other trade policies affects productivity, firm performance, and economic growth (Topalova and Khandelwal, 2011; Defever et al., 2020). However, most of this literature focuses on policies that reduce tariff barriers to trade. Less is known about the effects of relaxing non-tariff barriers, including those that are embedded within preferential trade agreements themselves.

I examine how relaxing non-tariff constraints embedded within trade agreements affects trade outcomes using the case of Jordan. In the late 1990s, Jordan entered into the Qualifying Industrial Zones (QIZ) arrangement with the United States and Israel. The QIZ agreement granted Jordanian producers operating within designated industrial zones duty-free access to the U.S. market conditional on satisfying specified input requirements. In particular, qualifying products were required to incorporate a minimum amount of inputs from Israel as part of the conditions under which U.S.-Israel preferential market access was extended to these industrial zones in Jordan. Regardless of the political motivations underlying the agreement, access to the tariff preference was therefore conditional on potentially costly non-tariff requirements, including restrictions on input sourcing and the location of production.

In the early 2000s, following bilateral negotiations between the United States and Jordan, the U.S.-Jordan Free Trade Agreement (FTA) introduced an alternative route to preferential access to the U.S. market. The agreement began phasing in tariff reductions in 2001 and was fully implemented by 2010. Although the FTA also contains rules-of-origin requirements, these requirements were less restrictive than those under the QIZ arrangement. In particular, the FTA did not require the use of Israeli inputs. Moreover, firms could produce anywhere in Jordan and export under the FTA rather than being required to locate production within a designated industrial zone. The FTA therefore provided an alternative route to preferential access without the QIZ-specific input-sourcing and locational requirements.

In this paper, I exploit the gradual introduction of the U.S.-Jordan FTA to examine how access to a less restrictive alternative to the QIZ program affects trade outcomes. The setting allows me to study what happens when exporters gain access to an alternative preferential regime that ultimately provides the same tariff-free access to the U.S. market while making the conditions required to obtain that access easier to satisfy.

To guide the empirical analysis, I develop a simple conceptual framework. An exporter chooses whether to export a product to the United States and, conditional on exporting, whether to enter under MFN treatment, the QIZ program, or the FTA. Preferential treatment under the QIZ or the FTA is attractive because it allows exporters to avoid tariffs, but claiming the preference requires them to incur other non-tariff costs to be eligible. The framework therefore characterizes the take-up of a preference regime by comparing the value of the tariff savings with the cost of satisfying the relevant eligibility requirements. Because the FTA imposes less restrictive requirements than the QIZ program, it lowers the cost of obtaining preferential treatment and reduces the threshold at which claiming a preference becomes worthwhile.

The model delivers three main hypotheses. First, as scheduled FTA tariffs decline, products already exported under the QIZ program should increasingly switch toward the FTA.

Second, as FTA liberalization lowers the cost of accessing the U.S. market, U.S. exports should increase relative to exports to other destinations, particularly in industries with a strong pre-existing presence in the U.S. market. Third, as FTA liberalization increases the profitability of serving the U.S. market, products receiving larger tariff reductions should become more likely to enter the U.S. market.

To test these predictions, I construct a product-level dataset combining information from several sources. First, for the empirical analysis, I use U.S. customs data on imports from Jordan at the eight-digit product level from 1998 to 2022. These data identify the trade program under which each product enters the United States, allowing me to distinguish imports receiving preferential treatment under the QIZ or U.S.-Jordan FTA from those entering under MFN treatment or other programs. Second, I digitize the U.S.-Jordan FTA tariff schedule, including the product-specific phaseout of tariffs between 2001 and 2010. Third, I use UNCTAD data on Jordanian exports by product and destination to measure broader product-level export activity. Finally, I obtain U.S. MFN tariff rates from the WTO database. U.S. customs data are also used to study product entry into the U.S. market.

I use a set of complementary empirical strategies to test the mechanisms highlighted by the model. I begin by examining whether products already using the QIZ program shift toward the FTA as preferential access under the FTA becomes more valuable. I exploit cross-product variation in the timing and magnitude of the negotiated FTA tariff phaseout. The sample is restricted to products observed entering the United States under the QIZ program before their scheduled FTA liberalization begins, and I follow these products throughout the phase-in period. I estimate using both OLS and a stacked PPML specification in which each product-year is observed separately under the FTA, QIZ, no-preference, and other programs. Product-by-year fixed effects absorb changes in total U.S. imports of a product, while product-by-program and program-by-year fixed effects absorb persistent differences in program use and common changes in program utilization over time. The coefficient of interest therefore captures whether larger scheduled FTA tariff concessions reallocate a product's U.S. imports from the QIZ program toward the FTA. Consistent with the model, I find a large and statistically significant shift from QIZ toward FTA utilization following liberalization.

I next examine export growth among products treated by the FTA. In particular, I ask whether the reduction in trade barriers generated by the agreement led to stronger export growth in industries that were already well positioned to serve the U.S. market. To do so, I exploit variation in the timing of tariff elimination across eight-digit products between 2001 and 2010. I estimate an event-study specification in which the event is the year in which the FTA tariff on a product reaches zero, and compare the subsequent growth of exports to the United States to non-U.S. destinations. I interact the event-time indicators with the pre-FTA share of exports to the United States at the two-digit industry level, thereby testing whether products in industries with a stronger presence in the U.S. market experience larger export responses. I find that, once the FTA tariff reaches zero, exports to the United States increase substantially relative to exports to non-U.S. destinations in industries with higher presence in the U.S. market, and this increase persists over time.

Finally, I examine the extensive margin of trade. The model predicts that reducing the cost of accessing the U.S. market can make exporting profitable for products that were not previously sold there. I test this prediction by exploiting differences across products in

their potential gains from tariff liberalization, which depend on their baseline tariff exposure. Specifically, I examine whether products facing larger tariff reductions are more likely to enter the U.S. market following liberalization. Consistent with the model, I find that products with greater potential tariff savings are significantly more likely to begin exporting to the United States after the FTA is implemented.

Taken together, these results show that relaxing the non-tariff barriers attached to preferential trade agreements affects trade along several distinct margins. The introduction of the U.S.-Jordan FTA induced exporters to substitute away from the more restrictive QIZ regime, increased exports of products in industries with a stronger pre-existing presence in the U.S. market, and promoted entry of new products into the U.S. market. More broadly, the findings suggest that the economic effects of preferential trade agreements depend not only on the tariffs they eliminate but also on the non-tariff costs products must incur to qualify for preferential treatment.

Related Literature

This paper contributes to a growing literature on preferential trade agreements, preference utilization, and rules of origin. A central insight of this literature is that the value of preferential market access depends not only on the size of the tariff preference but also on the costs exporters must incur to qualify for it. Rules of origin, documentation requirements, and other eligibility conditions can impose administrative costs and restrict firms' sourcing decisions, reducing the effective value of preferential tariffs and leading some exporters not to claim preferences even when they are formally available (Anson et al., 2005; Cadot et al., 2006; Manchin, 2006; Keck and Lendle, 2012; Hakobyan, 2015). Consistent with this mechanism, the use of trade preferences tends to rise the larger the tariff savings, making it more worthwhile for exporters to incur the costs associated with satisfying eligibility requirements. I build on this literature by studying what happens when exporters obtain access to an alternative preferential regime that relaxes these requirements while preserving preferential access to the same destination market.

The paper is especially closely related to work on rules of origin and input-sourcing restrictions. Rules of origin determine which products qualify as originating within a preferential trade area and can therefore receive preferential tariff treatment. Although these requirements are intended to prevent trade deflection, they can themselves distort trade by inducing firms to alter where they source intermediate inputs. Conconi et al. (2018), for example, show that NAFTA rules of origin reduced imports of intermediate goods from non-member countries relative to imports from NAFTA partners, demonstrating that rules of origin can effectively extend protection from final goods to inputs. More recent preliminary work by Conconi et al. (n.d.) provides firm-level evidence on the trade-distorting effects of rules of origin, further emphasizing that these requirements shape firms' sourcing and trade decisions. This mechanism is particularly transparent in the Jordanian setting, where access to QIZ preferences explicitly required producers to incorporate inputs from Israel. The QIZ arrangement therefore provides a setting in which preferential market access was directly tied to a politically specified sourcing requirement.

This paper is also closely related to Sytsma (2021) and Sytsma (2022), who study the consequences of relaxing rules of origin under the European Union's Generalized System of

Preferences. The reform they examine allowed apparel producers in least-developed countries to use a broader set of internationally sourced textile inputs while continuing to receive preferential access to the European market. Sytsma (2021) shows that restrictive rules of origin significantly reduced the utilization of available trade preferences, while Sytsma (2022), using transaction-level data from Bangladeshi apparel exporters, finds that relaxing the sourcing requirements substantially increased firm-level exports. His estimates imply that the original rules of origin sharply reduced the effective value of the tariff preference, illustrating how non-tariff eligibility conditions can offset a substantial portion of the gains from trade agreements.

The Jordanian setting differs from these studies in an important way. Rather than changing the rules of origin within an existing preferential regime, the U.S.-Jordan FTA introduced a second route to preferential access alongside the QIZ program. Under QIZ, qualifying exporters could already obtain duty-free access to the U.S. market, but only if they satisfied Israeli input requirements and produced within designated industrial zones. As the FTA was gradually implemented, exporters obtained an alternative route to preferential treatment that did not require Israeli inputs and did not restrict production to designated zones. The setting therefore allows me to observe the response to a relaxation of preferential eligibility requirements through exporters' endogenous choice across overlapping trade regimes.

This feature allows me to distinguish between several margins that are difficult to separate in settings where a reform simultaneously changes both tariffs and rules of origin for all exporters. For products already exporting through QIZ, access to the FTA primarily relaxes the non-tariff requirements associated with obtaining preferential access because these products had already entered the U.S. market duty free. A shift from QIZ to the FTA can therefore occur even in the absence of a substantial change in total exports. By contrast, products initially entering under MFN treatment gain both a tariff reduction and access to a less restrictive preferential regime. Finally, sufficiently large reductions in the overall cost of serving the U.S. market may induce products that were previously not exported to the United States to enter. The paper thus studies how relaxing preferential eligibility requirements affects regime choice, preference utilization, the intensive margin of trade, and product entry.

The paper also relates more broadly to the literature studying how trade agreements affect trade beyond their statutory tariff provisions. Preferential agreements frequently contain complex provisions governing origin, sourcing, documentation, and production processes, and a growing body of evidence shows that these provisions can substantially alter their effective degree of liberalization. Existing work documents that restrictive rules of origin can reduce trade flows and preference utilization (Augier et al., 2005; Anson et al., 2005; Cadot et al., 2006), while other studies show that differences in administrative and compliance costs help explain why exporters choose among overlapping preferential regimes (Bureau et al., 2007; Keck and Lendle, 2012). The Jordanian case provides a clean setting for examining these forces because the QIZ and FTA offered exporters alternative routes to preferential access to the same market but differed sharply in the conditions that had to be satisfied to obtain that access.

My contribution to this literature is therefore threefold. First, I provide evidence on how exporters reallocate across trade regimes when a less restrictive alternative becomes available.

Second, I show that the introduction of a less restrictive trade agreement may increase exports among industries with a large presence in the destination market. Third, I study the extensive margin and show that reducing the non-tariff costs of obtaining preferential access induces new products to enter the U.S. market.

More broadly, the findings complement the literature on trade liberalization by emphasizing that liberalization can occur not only through reductions in tariffs but also through reductions in non-tariff barriers. Much of the empirical literature on trade policy identifies the effects of changes in tariffs. The evidence in this paper instead highlights the importance of the institutional architecture surrounding preferential access. Two agreements can ultimately provide similar tariff treatment while generating different trade outcomes if the costs of satisfying their eligibility requirements differ substantially. Understanding the effects of preferential trade agreements therefore requires accounting not only for the preferences they offer, but also for the restrictions exporters must satisfy in order to use them.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting. Section 3 develops a conceptual framework and derives the main empirical hypotheses. Section 4 describes the data and empirical strategies used to test these hypotheses. Section 5 presents the empirical results. Section 6 concludes.

2 Institutional Setting

During the period studied in this paper, Jordanian exporters could obtain preferential access to the U.S. market through two distinct regimes: the Qualifying Industrial Zones (QIZ) program and, starting 2001, the U.S.–Jordan Free Trade Agreement (FTA). Both programs provide a route to duty-free access, but they differ substantially in the conditions firms had to satisfy to use them. In particular, the FTA removed two constraints embedded in the QIZ agreement: in order to access the U.S. market tariff-free, exporters under the QIZ program had to produce within a designated industrial zone and to source a fixed share of the product's value from Israel.

The QIZ program emerged from U.S. efforts to deepen economic cooperation between Israel and its Arab neighbors following the peace agreements of the 1990s. Jordan and Israel signed a peace treaty in 1994, and the United States subsequently established the legal framework for Qualifying Industrial Zones in 1996. The first Jordanian QIZ was designated in Jordan in 1997, with additional zones added thereafter (Congressional Research Service, 2013; U.S. Department of Commerce, n.d.-b). The program was followed by a rapid expansion of Jordanian exports to the United States, particularly in textiles and apparel (Saif, 2006; Pelzman, 2011).

Access to QIZ preferences, however, was conditional on both the location of production and the sourcing of inputs. Qualifying goods had to be produced or substantially transformed within an officially designated QIZ and satisfy a 35 percent content requirement. For Jordanian exports, at least 11.7 percent of the product's appraised value had to consist of Jordanian content and at least 8 percent had to originate in Israel. The remaining 15.3 percent could come from a combination of a Jordanian QIZ, Israel, the United States, and the West Bank and Gaza (Congressional Research Service, 2013; U.S. Department of Commerce, n.d.-b). Thus, an exporter could not obtain QIZ treatment simply by undertaking sufficient

production in Jordan. It also had to locate qualifying production inside a designated zone and ensure that a fixed share of the product's value originated in Israel.

The U.S.–Jordan FTA introduced an alternative set of conditions for obtaining preferential access. The agreement was signed on October 24, 2000, entered into force on December 17, 2001, and gradually eliminated U.S. tariffs on Jordanian products, until full implementation of tariff removal in 2010 (Office of the U.S. Trade Representative, n.d.,n). Unlike QIZ treatment, FTA preferences were available for qualifying production throughout Jordan rather than only within designated industrial zones.

The FTA also changed the composition of the rules-of-origin requirement. Like QIZ, the FTA stipulates a 35 percent qualifying-content threshold. Under the FTA, however, the threshold could be satisfied by the value of materials produced in Jordan and direct processing costs incurred in Jordan, while U.S.-origin materials could account for up to 15 percentage points of these required 35 percent (United States and Jordan, 2000). Thus, if the full U.S. allowance is used, at least 20 percentage points must be accounted for by Jordanian materials and processing. Most importantly, unlike QIZ, the FTA did not require any minimum share of the product to originate in Israel.

Table A1 in the Appendix summarizes the eligibility requirements under the two regimes.

The distinction is important because the FTA did not lower the headline 35 percent content requirement. Rather, it changed how firms could satisfy that requirement. Under QIZ, 8 percentage points were tied necessarily to a particular foreign source, Israel. This requirement could constrain firms' ability to choose input suppliers to minimize their costs. Under the FTA, mandatory Israeli sourcing disappeared. For firms able to satisfy the Jordanian production requirement, the FTA therefore provided greater flexibility in choosing where to source intermediate inputs. Moreover, the FTA also relaxed the geographic restriction because FTA exports were not required to be produced in a designated area.

The attractiveness of the FTA increased gradually because tariff reductions were phased in at different rates across products. Figure 1 shows this process. By 2002, 69.2 percent of covered HTS8 product lines were duty free under the FTA, but these products accounted for only 47.0 percent of Jordan's pre-FTA exports to the United States. By 2005, 97.6 percent of product lines had reached zero tariffs, while their import-weighted share was only 71.1 percent. The remaining 2.4 percent of product lines accounted for approximately 28.9 percent of pre-FTA import value and were not fully liberalized until 2010. Liberalization therefore proceeded more rapidly in terms of the number of products than in terms of their economic importance.

The gradual expansion of FTA access coincided with a pronounced shift in the program under which Jordanian goods entered the United States. Figure 2 plots total U.S. imports from Jordan together with imports entering under QIZ and FTA treatment. Imports grew rapidly after 2000 and initially entered predominantly through QIZ. QIZ imports reached approximately $1 billion in the mid-2000s before declining substantially between 2007 and 2011. Over the same period, imports entering under the FTA increased and overtook QIZ imports around 2010, when tariff liberalization under the agreement was fully implemented.

Following full implementation, nearly all subsequent growth in U.S. imports from Jordan was recorded under the FTA. Total imports increased from approximately $1 billion in 2010 to $3.38 billion in 2024, of which $2.92 billion entered under the FTA. QIZ imports remained positive after 2011 but represented a very small share of total exports. Appendix

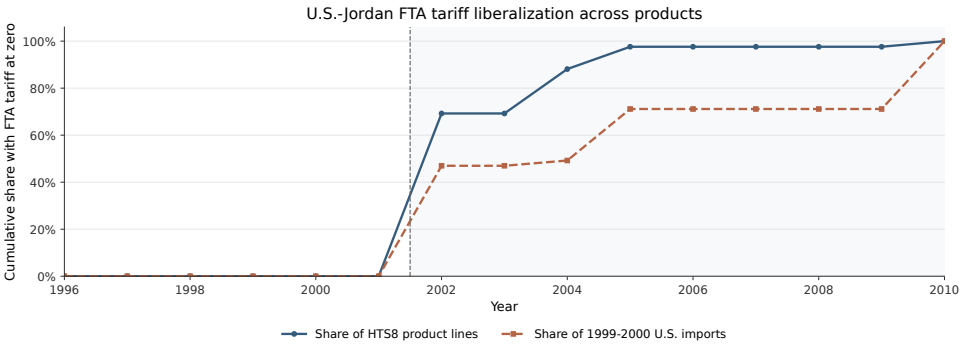


Figure 1: Tariff Liberalization under the U.S.–Jordan FTA

Notes: The product-line denominator is a fixed universe of 9,374 verified base-free or clean standard HTS8 schedule lines. Import weights are fixed 1999–2000 U.S. customs values among exact schedule-matched lines, covering 98.4 percent of total 1999–2000 imports. Because 2001 was a partial implementation year, lines legally at zero in 2001 first count in 2002.

Figure A1 shows that Jordanian imports from Israel also declined at the same time U.S. imports from Jordan shifted away from QIZ and toward the FTA. This pattern is consistent with exporters reducing reliance on the QIZ regime, and and therefore no longer needing to satisfy its Israeli-input requirement, as they switched to the FTA.

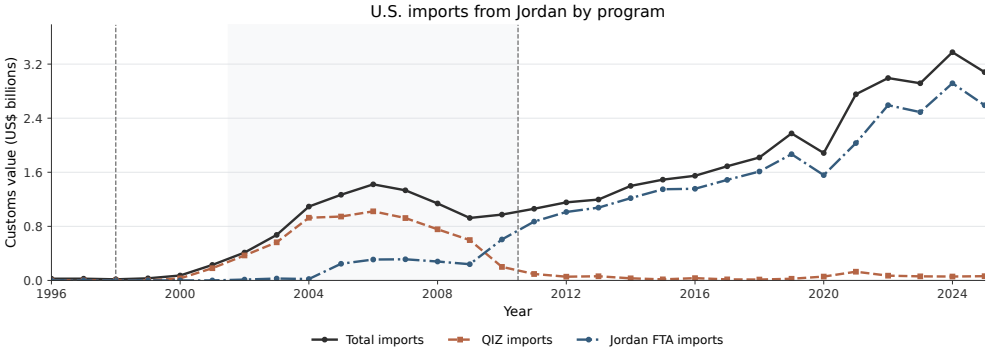


Figure 2: U.S. Imports from Jordan by Trade Program

Notes: Annual U.S. imports for consumption from Jordan, measured by customs value. Total imports include QIZ, Jordan FTA, no-preference, and other-program entries. The shaded area marks the FTA tariff phase-in period; 2001 was a partial implementation year.

Therefore, the introduction of the U.S.-Jordan FTA altered the conditions under which Jordanian exporters obtained preferential access to the U.S. market. QIZ provided duty-free access subject to both a designated-zone requirement and mandatory Israeli sourcing. The FTA also embedded a substantial origin requirement but removed both of these QIZ-specific constraints, providing firms with greater flexibility over where to produce and from whom to source inputs.

3 Conceptual Framework

This section develops a simple framework that organizes the empirical analysis around three margins of adjustment to the U.S.–Jordan Free Trade Agreement (FTA): the choice of preferential trade regime, the expansion of existing exports, and the entry of new products into the U.S. market.

3.1 Export profits and the choice of trade regime

Consider a Jordanian product h that can potentially be exported to the United States. Let q denote the quantity sold in the U.S. market, and let $R_h(q, a_h)$ denote revenue from U.S. sales. The parameter a_h captures the product's underlying ability to serve the U.S. market, including factors such as established relationships with U.S. buyers, distribution networks, obtained U.S. certifications, and production capacity oriented toward the U.S. market.

Let $C_h(q)$ denote the cost of producing and exporting quantity q, excluding costs that arise specifically from the trade regime used to enter the United States.

I summarize the tariff and non-tariff burden associated with each regime by an ad valorem cost m_{ht}^r, where $r \in \{M, Q, F\}$ denotes MFN, QIZ, and FTA treatment, respectively. These costs are

$$\begin{aligned} m_h^M &= t_h^M, \\ m_h^Q &= c_h^Q, \\ m_{ht}^F &= \tau_{ht}^F + c_h^F. \end{aligned}$$

Here, t_h^M is the MFN tariff, τ_{ht}^F is the FTA tariff applied to product h in year t, and c_h^Q and c_h^F summarize the additional costs associated with satisfying the requirements of the QIZ and the FTA, respectively.

Because QIZ imposes sourcing and production-location requirements that do not apply in the same form under the FTA, I assume that

$$c_h^F < c_h^Q.$$

For simplicity, I assume that the U.S. tariff is fully reflected in the net return received by the Jordanian exporter. Legally, the tariff is paid at the U.S. border by the importer. The framework only requires that a higher U.S. tariff reduces the profitability of exporting from Jordan to the United States.

Under regime r, the exporter chooses the quantity that maximizes profits:

$$\pi_{ht}^r = \max_{q \geq 0} \left\{ (1 - m_{ht}^r) R_h(q, a_h) - C_h(q) \right\}. \tag{1}$$

Conditional on exporting to the United States, let $\mathcal{R}_{ht}$ denote the set of available regimes. The exporter chooses the regime that generates the highest profit:

$$r_{ht}^* = \arg\max_{r \in \mathcal{R}_{ht}} \pi_{ht}^r. \tag{2}$$

Let $q_h^*(m, a_h)$ denote the optimal quantity exported at market-access cost m. By the envelope theorem, profits decline with the market-access cost:

$$\frac{\partial \pi_h(m, a_h)}{\partial m} = -R_h\left(q_h^*(m, a_h), a_h\right) < 0. \tag{3}$$

Under standard assumptions on the revenue and cost functions, the optimal quantity exported declines with the cost of accessing the U.S. market:

$$\frac{\partial q_h^*(m, a_h)}{\partial m} < 0. \tag{4}$$

Thus, a reduction in the effective cost of accessing the U.S. market increases both the profitability of exporting and the optimal quantity exported.

These two relationships generate the three hypotheses below.

3.2 Hypothesis 1: Reallocation toward the FTA

Consider first products that are already exported to the United States. The introduction of the FTA creates a new alternative to MFN and QIZ treatment.

For a product eligible for QIZ, the effective cost of QIZ treatment is

$$m_h^Q = c_h^Q, \tag{5}$$

whereas the effective cost of FTA treatment is

$$m_{ht}^F = \tau_{ht}^F + c_h^F. \tag{6}$$

QIZ provides duty-free access throughout the period, whereas the FTA tariff remains positive for many products during the phase-in. QIZ can therefore initially remain attractive despite its more restrictive requirements.

As the scheduled FTA tariff declines, however, the effective cost of FTA treatment falls:

$$\frac{\partial m_{ht}^F}{\partial \tau_{ht}^F} = 1 > 0. \tag{7}$$

Together with equation (3), this implies that declining FTA tariffs raise the profitability of FTA treatment relative to QIZ.

The comparison becomes particularly transparent when the FTA tariff reaches zero. At that point,

$$m_h^F = c_h^F < c_h^Q = m_h^Q. \tag{8}$$

Both programs then provide duty-free access, but the FTA does so without the additional QIZ sourcing and production-location requirements. Exporters eligible for both programs therefore have an incentive to shift existing trade from QIZ toward the FTA.

The same logic applies to MFN treatment: as the preferential FTA tariff declines relative to the MFN tariff, FTA treatment becomes increasingly attractive.

This yields the first hypothesis:

Hypothesis 1: FTA utilization. *As scheduled FTA tariffs decline, an increasing share of Jordanian exports to the United States enters under the FTA rather than under alternative trade regimes. In particular, products previously entering through QIZ should increasingly switch toward FTA treatment as FTA liberalization advances.*

This hypothesis concerns the regime under which existing trade enters the United States. Program switching can therefore occur without any change in the total value of exports.

3.3 Hypothesis 2: Expansion of U.S. exports

The second hypothesis concerns the quantity exported rather than the choice of trade regime.

Equation (4) implies that a decline in the effective cost of accessing the U.S. market increases exports. As FTA tariffs fall, the effective cost of FTA treatment declines. Once the FTA tariff reaches zero,

$$m_h^F = c_h^F < c_h^Q = m_h^Q. \tag{9}$$

At this point, exporters can obtain duty-free access through the FTA without facing the additional requirements associated with QIZ. Reaching zero therefore provides a natural empirical benchmark at which the FTA's tariff disadvantage relative to QIZ has been fully eliminated.

The size of the export response need not be identical across industries. Industries with a substantial presence in the U.S. market prior to the FTA are likely to possess complementary capabilities that allow them to respond more strongly to improved market access. These may include established U.S. buyers, distribution networks, market-specific knowledge, certifications, and production capacity already oriented toward U.S. demand.

The parameter a_h captures these pre-existing U.S.-market capabilities. I assume that

$$\frac{\partial q_h^*(m, a_h)}{\partial a_h} > 0,$$

and, more importantly, that the response to a reduction in market-access costs is stronger for products with greater U.S.-market capability:

$$\frac{\partial^2 q_h^*(m, a_h)}{\partial m \, \partial a_h} < 0. \tag{10}$$

This condition captures a complementarity between improved market access and pre-existing capacity to serve the U.S. market. Industries already oriented toward the United States should be able to expand more rapidly when the cost of accessing that market falls.

Moreover, the U.S.–Jordan FTA directly changes the cost of accessing the United States but does not directly reduce the cost of exporting Jordanian products to other destinations. The relevant comparison is therefore between exports to the United States and exports of the same products to non-U.S. markets.

This yields the second hypothesis:

Hypothesis 2: Export growth. *As FTA liberalization reduces the cost of accessing the U.S. market, exports to the United States increase relative to exports to non-U.S. destina-*

tions. This increase should be larger for products belonging to Jordanian industries with a stronger pre-existing presence in the U.S. market.

3.4 Hypothesis 3: Product entry into the U.S. market

The FTA can also affect the extensive margin by making it profitable to export products that were previously absent from the U.S. market.

Consider a product that is not initially sold in the United States. Entry occurs when the maximum profit available from serving the U.S. market becomes positive:

$$\max_{r \in \mathcal{R}_{ht}} \pi_{ht}^{r} > 0. \tag{11}$$

Because profits increase when market-access costs fall, FTA liberalization can cause products that were previously below this profitability threshold to enter the U.S. market.

The magnitude of this effect differs across products because the size of the negotiated tariff reduction differs across products. Let the potential tariff saving for product h be

$$S_h = \tau_h^{F,0} - \tau_h^{F,\text{final}}, \tag{12}$$

where $\tau_h^{F,0}$ is the tariff at the beginning of the FTA phase-in and $\tau_h^{F,\text{final}}$ is the tariff after liberalization is complete. For products whose FTA tariff is eventually eliminated,

$$S_h = \tau_h^{F,0}. \tag{13}$$

Products with larger potential tariff savings experience larger reductions in the cost of accessing the U.S. market. Equation (3) therefore implies a larger increase in the profitability of entry for these products.

This yields the third hypothesis:

Hypothesis 3: Product entry. *FTA liberalization increases the probability that products previously absent from the U.S. market begin exporting to the United States. The increase in entry probability should be larger for products receiving greater potential tariff reductions under the FTA.*

3.5 Summary

The framework generates three distinct but related effects of FTA liberalization.

First, the FTA changes the relative profitability of the available trade regimes. As FTA tariffs decline, existing U.S.-bound trade increasingly shifts from QIZ, MFN, and other regimes toward FTA treatment.

Second, FTA liberalization lowers the effective cost of serving the U.S. market. Once the FTA tariff reaches zero, exporters can obtain duty-free access without facing the additional sourcing and locational restrictions associated with QIZ. This generates an intensive-margin increase in U.S. exports, particularly in industries that already have a strong presence in the U.S. market.

Third, lower FTA tariffs increase the profitability of U.S. market entry. Products that were previously not profitable to export to the United States may enter as market-access costs decline, with larger responses among products receiving larger tariff reductions.

4 Data and Empirical Strategy

4.1 Data

The empirical analysis combines four sources of product-level data. First, I use U.S. customs data to observe imports from Jordan and the trade program under which those imports enter the United States. Second, I digitize the product-level tariff schedule of the U.S.–Jordan FTA. Third, I use U.S. MFN tariffs from the World Trade Organization (WTO) to measure the value of obtaining preferential access to the U.S. market. Finally, I use UNCTAD data on Jordanian exports by product and destination to study changes in export values.

These datasets serve different purposes. The U.S. customs data and the FTA tariff schedule are used to study switching across trade regimes and product entry. The UNCTAD data are used to study whether the FTA affected exports themselves. The WTO MFN tariff provides a predetermined measure of how valuable preferential U.S. access is for each product.

U.S. customs data. For the empirical analysis, I use U.S. customs data on imports from Jordan from 1998 to 2022 at the eight-digit Harmonized Tariff Schedule (HTS8) level. The data identify the customs value entering under different trade programs. I group these programs into four categories: the U.S.–Jordan FTA, QIZ, no-preference treatment, and other preferential programs.

Let

$$V_{kpt}$$

denote imports of product k under program p in year t, where

$$p \in \{\text{FTA}, \text{QIZ}, \text{MFN}, \text{Other}\}.$$

I refer to imports entering without a preference as MFN imports.

A product may use more than one program in the same year. The customs data therefore allow me to observe how a product's U.S. imports are allocated across QIZ, FTA, and non-preferential treatment. I use these data to study both switching from QIZ to the FTA and preference take-up among products that previously entered without preferences.

For some of the analysis, I classify products according to the program they used before implementation of the FTA. I use 1999–2000 as the pre-FTA period. Products entering under QIZ during this period are classified as pre-FTA QIZ products, while products entering without preferences are classified as pre-FTA MFN products. The classification is fixed using pre-FTA information and does not change according to subsequent FTA adoption.

FTA tariff schedule. I digitize the U.S. tariff schedule contained in the U.S.–Jordan FTA and construct the scheduled preferential tariff for each HTS8 product in each year.

Let

$$\tau_{kt}^{FTA}$$

denote the FTA tariff faced by product k in year t, and let

$$\tau_{k}^{base}$$

denote its negotiated base tariff. I measure cumulative tariff liberalization as

$$L_{kt} = \tau_{k}^{base} - \tau_{kt}^{FTA}. \tag{14}$$

A larger value of L_{kt} therefore corresponds to a larger FTA tariff concession. I also identify the first year in which each product receives any tariff reduction under the FTA. The product-level staging of these tariff reductions provides the main source of variation for the analysis of switching from QIZ toward the FTA.

WTO MFN tariffs. I obtain U.S. MFN tariffs from the WTO database. I use the tariff prevailing before implementation of the FTA and denote it by

$$MFN_k.$$

Destination-level export data. I use UNCTAD data on Jordanian exports by six-digit product, destination country, and year. Let

$$X_{kdt}$$

denote exports of product k to destination d in year t.

The destination dimension allows me to compare changes in exports to the United States with changes in exports of the same product to other markets. This is useful because Jordanian products may experience changes in production capacity, productivity, or global demand that have nothing to do with the U.S.–Jordan FTA. Observing the same product across destinations makes it possible to absorb these broader product-specific shocks.

Products are harmonized to a fixed HS6 classification over time. I merge the pre-FTA QIZ/MFN classification and the WTO MFN tariff onto the destination-level trade panel. Contemporaneous FTA utilization is not used as a treatment variable in the export regressions because the decision to use the FTA is itself an outcome of the policy.

For the destination-level regressions, I use UNCTAD data for both U.S. and non-U.S. exports so that treated and comparison destinations are measured using the same data source. U.S. customs imports from Jordan provide an alternative measure of U.S. trade with Jordan and are used for the exercises on trade regime utilization and product entry.

5 Empirical Strategies and Results

5.1 Hypothesis 1: Reallocation toward the FTA

The first hypothesis predicts that products previously exported under the QIZ program increasingly switch to the U.S.–Jordan FTA as the FTA tariff decreases and becomes more favorable. During the FTA's phase-in period, products faced a tradeoff between QIZ's tariff advantage and the FTA's less restrictive eligibility requirements. As FTA tariffs declined, this tariff disadvantage narrowed; once the FTA tariff reached zero, both programs provided equivalent tariff-free access, while the FTA did not impose the QIZ's input-sourcing and locational requirements. Greater substitution toward the FTA as its tariff disadvantage disappears is therefore consistent with exporters valuing the FTA's less restrictive eligibility conditions in addition to tariff preferences.

I begin by examining products with positive QIZ imports in either 1999 or 2000, before the FTA, and hold this cohort of products fixed over time. Figure 3 shows a substantial reallocation from QIZ to the FTA within this group. The FTA share of the cohort's total U.S. imports increased from 22 percent in 2009 to 62 percent in 2010 and 91 percent in 2011. By 2020, approximately 95 percent of imports entered under the FTA, compared with only 4 percent under QIZ. The timing of this reallocation is consistent with exporters shifting from QIZ to the FTA as the latter became increasingly attractive. This aggregate pattern is nevertheless only suggestive, because it cannot distinguish the effect of the product-specific FTA tariff schedule from changes in program use that are common across products. Appendix Figure A5 reports the corresponding series using equal product weights.

To test whether this reallocation is systematically related to the negotiated tariff schedule, I restrict attention to products with positive QIZ imports before their first scheduled FTA tariff cut. This product-specific sample contains 36 eight-digit products, corresponding to 29 six-digit products. I also restrict the sample to 1998–2010, since all products have a zero FTA tariff after 2010. I define cumulative FTA liberalization as

$$L_{ht}^{10} = \frac{\tau_h^{base} - \tau_{ht}^{FTA}}{10},$$

where one unit corresponds to a 10-percentage-point reduction in the FTA tariff relative to the base tariff specified in the agreement. Liberalization is set to zero during 1998–2001 and follows the product-specific tariff schedule from 2002 through 2010.

The preferred specification stacks FTA and QIZ customs values for each product-year and estimates

$$\mathbb{E}[V_{hpt}] = \exp\left[\alpha_{ht} + \mu_{hp} + \lambda_{pt} + \beta \mathbf{1}\{p = \text{FTA}\} * L_{ht}^{10}\right] \tag{15}$$

where $p \in \{\text{FTA}, \text{QIZ}\}$ is the program, with QIZ serving as the omitted program, h is the eight-digit product and t is time. Product-by-year fixed effects absorb shocks common to both programs within a product-year. Product-by-program fixed effects absorb persistent differences in the propensity of a product to use each program, while program-by-year fixed effects absorb aggregate changes in the relative use of FTA and QIZ over time. The coefficient β therefore captures whether products receiving larger FTA tariff concessions shift more

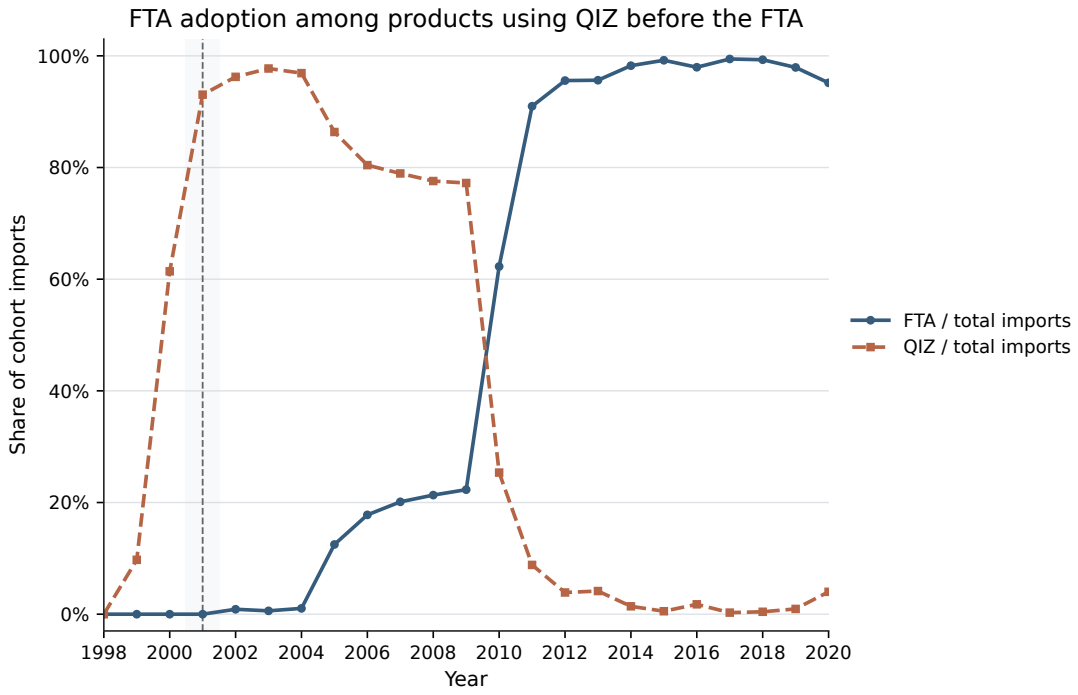


Figure 3: FTA Adoption among Products Using QIZ before the FTA

Notes: The cohort contains HTS8 products with positive QIZ imports in 1999 or 2000 and is held fixed thereafter. The figure reports aggregate FTA and QIZ customs values as shares of total U.S. imports for the cohort. The shaded area marks the partial implementation year 2001.

strongly toward the FTA relative to QIZ than other products in the same year. The model is estimated by PPML, which accommodates zero program flows (Santos Silva and Tenreyro, 2006), with standard errors clustered by HTS8 product.

As a complementary specification, I estimate an OLS regression using the FTA share of preferential imports,

$$\frac{FTA_{ht}}{FTA_{ht} + QIZ_{ht}},$$

as the dependent variable. The FTA share is regressed on cumulative tariff liberalization with product and year fixed effects. The regression is weighted by contemporaneous FTA-plus-QIZ customs value, placing greater weight on economically important preferential trade flows.

Table 1: Scheduled FTA Liberalization and Switching from QIZ to FTA

	(1) Value-weighted share	(2) FTA–QIZ PPML	(3) All-program PPML
FTA liberalization (10 pp)	0.097**		
	(0.046)		
FTA × liberalization (10 pp)		1.999*	1.904*
		(1.098)	(1.058)
MFN × liberalization (10 pp)			0.396**
			(0.196)
HTS8 and year fixed effects	Yes	No	No
HTS8 × year fixed effects	No	Yes	Yes
HTS8 × program fixed effects	No	Yes	Yes
Program × year fixed effects	No	Yes	Yes
Risk-set HTS8 lines	36	36	36
Harmonized HS6 products	29	29	29
Input observations	315	936	1872
Estimation observations	314	559	929

Notes: The product cohort contains HTS8 products with positive QIZ imports in at least one year strictly before the product's first scheduled FTA tariff cut. The panel spans 1998–2010; tariff liberalization is zero in 1998–2001 and equals the negotiated base tariff minus the exact scheduled FTA tariff in 2002–2010. Column (1) uses the FTA share of FTA plus QIZ customs value and weights by contemporaneous FTA plus QIZ value. Columns (2) and (3) stack program-specific customs values and retain zero flows; QIZ is the omitted program. Column (3) also includes MFN program cells. Standard errors in parentheses are clustered by HTS8. HS6-clustered PPML p-values are 0.075 in column (2) and 0.078 in column (3). Separation removes cells without identifying outcome variation, including all pre-FTA FTA cells. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$ using two-sided tests.

Table 1 provides evidence consistent with Hypothesis 1. In the value-weighted share OLS estimation, the coefficient on cumulative liberalization is 0.097 and statistically significant at the 5 percent level. Because L_{ht}^{10} is measured in 10-percentage-point tariff concessions, the estimate implies that a 10-percentage-point larger FTA tariff concession is associated with a 9.7-percentage-point increase in the FTA share of FTA-plus-QIZ U.S. imports.

The preferred FTA-versus-QIZ PPML specification yields a coefficient of 1.999 with a

standard error of 1.098 ($p = 0.069$). Although estimated less precisely, the coefficient is positive and economically sizable, indicating that products experiencing greater FTA liberalization shift toward FTA treatment relative to QIZ. The all-program PPML specification yields a similar coefficient of 1.904 for FTA imports relative to QIZ imports. MFN imports also increase relative to QIZ as FTA liberalization progresses. Because QIZ is the omitted program, both positive coefficients partly reflect declining QIZ use. However, the coefficient for the FTA is substantially larger than the coefficient for MFN, indicating that the reallocation is considerably stronger toward the FTA. Importantly, these estimates identify changes in the allocation of imports across customs programs rather than increases in total U.S. imports of the affected products.

I next ask whether the same pattern appears more broadly when FTA use is compared with all non-FTA channels rather than only with QIZ. This exercise uses all HTS8 products with a clean FTA tariff schedule and defines non-FTA imports as imports entering under QIZ, MFN/no-preference, or other programs. In this specification I use all years from 1998–2022. I center event time on the year before each product reaches a zero FTA tariff and estimate the event study model. Because the outcome uses a $\log(1 + y)$ transformation, I interpret its magnitude with caution when zero flows are present (Chen and Roth, 2024):

$$\log(1+FTA_{pt}^{imports})-\log(1+NonFTA_{pt}^{imports}) = \alpha_h+\lambda_t+\theta_{HS6(h)}t+\sum_{k\neq 0}\beta_k \mathbf{1}\{event_{ht}=k\}+\varepsilon_{ht}. \tag{16}$$

The coefficients β_k trace changes in FTA imports relative to non-FTA imports at event time k, relative to the year immediately preceding the product's attainment of a zero FTA tariff. HTS8 fixed effects absorb persistent differences in FTA use across products, year fixed effects absorb aggregate changes in program utilization, and HS6-specific linear trends allow closely related products to follow different underlying trends in relative program use.

Figure 4 reports the corresponding TWFE event study estimates. The coefficients are close to zero immediately before products reach tariff-free treatment. Since the identification relies on the parallel trends assumption, the absence of clear pre-trends is reassuring. After event time 0, the coefficients become positive, indicating an increase in FTA use relative to non-FTA following the attainment of a zero FTA tariff. Because products reach zero FTA tariffs in different years, conventional two-way fixed-effects event-study estimates may generate false comparisons. I therefore report results from a Sun–Abraham cohort-interaction specification in the appendix, following Sun and Abraham (2021), which addresses this issue. Appendix Figure A6 shows a similar post-event increase in the Sun–Abraham specification.

Overall, the evidence supports the first margin of adjustment predicted by Hypothesis 1: as the FTA became more favorable relative to QIZ, products that had previously relied on QIZ increasingly shifted their U.S. imports toward the FTA. The reallocation is also particularly visible after products attain zero FTA tariffs. At that point, the FTA and QIZ provide equivalent tariff-free access, while the FTA does not impose the QIZ's additional input-sourcing and locational requirements. The post-zero shift is therefore consistent with exporters preferring the less restrictive FTA once QIZ no longer offers a tariff advantage.

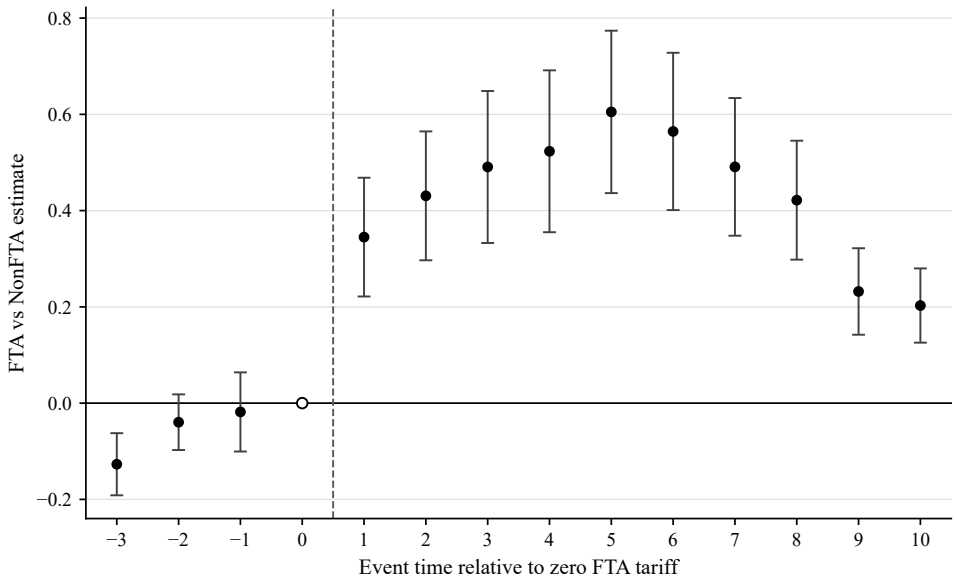


Figure 4: FTA Take-Up Relative to Non-FTA Imports Around Zero Tariffs

Notes: The figure reports TWFE event-study estimates using HTS8-year observations. The outcome is $\log(1+FTA)-\log(1+NonFTA)$, where non-FTA equals QIZ plus MFN/no-preference plus other program imports. Event time 0 is the year before the product reaches a zero FTA tariff. The specification includes HTS8 fixed effects, year fixed effects, and HS6-specific linear trends. Standard errors are clustered by HTS8. Bars report 95 percent confidence intervals.

5.2 Hypothesis 2: Expansion of U.S. exports

Hypothesis 1 shows that the gradual introduction of the FTA induced switching from other trade programs toward the FTA for exports to the United States. Hypothesis 2 asks whether this change was also associated with higher exports. The mechanism is not expected to operate equally across all products. Relaxing the QIZ sourcing and location requirements should matter most where Jordan already had the capacity needed to serve the U.S. market, as shown in the conceptual framework section. I therefore measure this pre-existing U.S. orientation at the HS2 level using each sector's share of Jordan's U.S. exports in 2000, before the FTA phaseout began.

Figure 5 motivates this exposure measure. U.S. imports from Jordan under the FTA after the full implementation in 2010 are concentrated in a small number of sectors, especially apparel and jewelry. These are also sectors that were economically important in Jordan's U.S. exports before the FTA became fully liberalized. The figure suggests that the relevant margin may not be an average effect across all products, but whether products in sectors with stronger pre-existing U.S. market access expanded more toward the United States as the FTA was liberalized.

To test this hypothesis, the sample contains clean HS6 products from UNCTAD with an identified FTA tariff schedule. I compare exports to the United States with exports of the same products to non-U.S. destinations, which are not directly treated by the U.S.–Jordan FTA. The event is the year in which all clean HTS8 lines within an HS6 product reach a zero

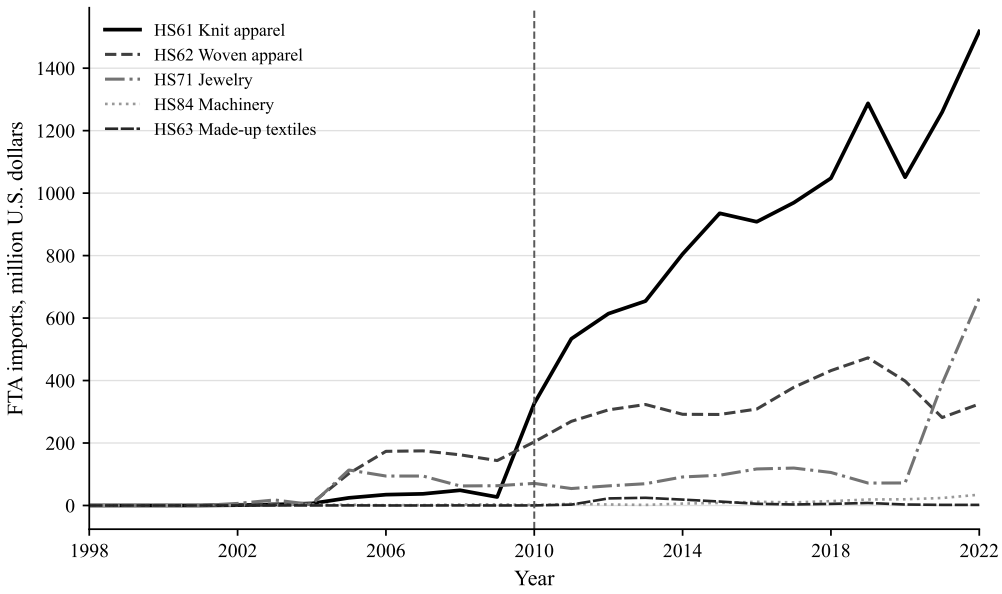


Figure 5: U.S. FTA Imports from Jordan by HS2 Sector

Notes: The figure reports annual U.S. imports from Jordan entering under the U.S.–Jordan FTA for the five HS2 sectors with the largest cumulative FTA imports during 2010–2022. Values are in millions of current U.S. dollars. The vertical dashed line marks 2010.

FTA tariff. At that point, the FTA and QIZ provide the same statutory tariff treatment, while the FTA does not impose the QIZ-specific sourcing and location requirements. I define event time 0 as the year before the zero-tariff year and event time 1 as the first year of zero-tariff treatment. Because treatment timing is staggered across products, I use the cohort-interaction event-study approach of Sun and Abraham (2021) to avoid comparisons between newly treated and already treated products.

The main specification is the event study below. Because the outcome uses a $\log(1+y)$ transformation, I interpret its magnitude with caution when zero flows are present (Chen and Roth, 2024). I also report a PPML robustness specification that preserves zero trade flows (Santos Silva and Tenreyro, 2006):

$$\log(1+X_{hdt}) = \alpha_{hd} + \delta_{ht} + \lambda_{dt} + \sum_{k\neq 0} \beta_k \, \mathbf{1}\{d = US\} \times \mathbf{1}\{event_{ht} = k\} \times HS2Share_{s(h)}^{2000} + \varepsilon_{hdt}. \tag{17}$$

Here X_{hdt} is Jordanian exports of HS6 product h to destination d in year t. The destination groups are the United States and the rest of the world. $HS2Share_{s(h)}^{2000}$ is the HS2 sector's share of Jordan's U.S. exports in 2000, measured in 10-percentage-point units. The fixed effects are demanding: HS6-by-destination fixed effects absorb persistent differences in product-destination trade; HS6-by-year fixed effects absorb product-specific shocks common across destinations; and destination-by-year fixed effects absorb aggregate U.S. versus non-U.S. demand conditions. Standard errors are clustered by HS6 product.

Figure 6 and Table A2 show a relative increase in exports to the United States after products reach zero FTA tariffs, concentrated among sectors with stronger pre-existing U.S.

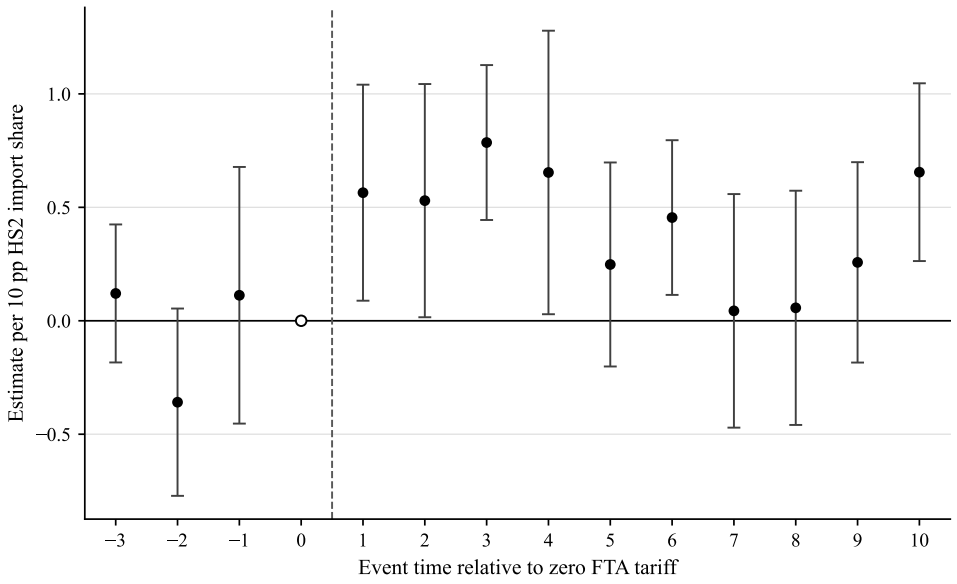


Figure 6: Export Growth Around Zero FTA Tariffs

Notes: The figure reports coefficients from equation (17). The outcome is $\log(1 + exports)$. Coefficients are measured per 10-percentage-point increase in the HS2 sector's 2000 share of Jordan's U.S. exports. Event time 0 is the year before the product reaches a zero FTA tariff; event time 1 is the first zero-tariff year. Bars report 95 percent confidence intervals.

market access. The estimates are small and statistically indistinguishable from zero in the three years before treatment. They become positive in the first zero-tariff year and remain positive for several years. At event time 1, a 10-percentage-point larger HS2 import share in 2000 is associated with a 0.565 log-point larger increase in exports to the United States relative to non-U.S. destinations. The corresponding estimates are 0.529 at event time 2 and 0.786 at event time 3. Appendix Figures A2 and A3 show that the same qualitative pattern appears in conventional TWFE and PPML event-study specifications.

I interpret these estimates as evidence that FTA liberalization was associated with relative export growth to the United States in sectors where Jordan already had U.S.-oriented export capacity. This does not imply that the FTA generated uniform export growth across all products. Nor does it by itself distinguish export expansion from destination reallocation. The identifying comparison is U.S. exports relative to non-U.S. exports of the same HS6 products; a positive coefficient can arise either because total exports expanded or because exports were redirected from non-U.S. destinations toward the U.S. market. I examine this in Appendix Figure A4, which estimates the effect of reaching a zero FTA tariff separately for U.S. exports, non-U.S. exports, and total exports. Because the unadjusted allocation exercise displays pre-treatment differences, the appendix reports the specification controlling for HS6-product specific linear trends. The trend-adjusted estimates are more consistent with export expansion than pure reallocation: U.S. exports and total exports rise after liberalization, while non-U.S. exports do not systematically fall in the early post-treatment years.

5.3 Hypothesis 3: FTA liberalization induces new product entry into the U.S. market

The final hypothesis examines the extensive margin. The FTA provides a less restrictive route to preferential access than QIZ, and this route becomes more valuable as the FTA tariff is phased out. Products that Jordan did not previously export to the United States should therefore become more likely to enter as FTA liberalization advances. I test this using U.S. customs data at the HTS8 product-year level. The sample consists of products with zero U.S. imports from Jordan in 1998–2000 and a clean FTA tariff schedule. The main outcome is first entry, defined as the first year in which U.S. imports of the product exceed $10,000. As a robustness check, I also use a lower $5,000 threshold to verify that the results are not driven by the particular cutoff used to define economically meaningful entry.

The estimating equation is

$$Entry_{ht} = \beta L_{ht}^{10} + \alpha_h + \delta_{st} + \varepsilon_{ht}, \tag{18}$$

where $Entry_{ht}$ is either the first-entry indicator or an annual indicator for imports above the threshold, L_{ht}^{10} is cumulative FTA tariff liberalization divided by 10, α_h are HTS8 product fixed effects, and δ_{st} denotes year, HS2-by-year, or HS4-by-year fixed effects. The preferred specification includes HS4-by-year fixed effects, so identification comes from differences in FTA liberalization across HTS8 products within the same HS4 sector and year. Standard errors are clustered by HTS8 product.

Table 2: FTA Tariff Liberalization and New U.S. Product Entry

Outcome	Fixed effects	$10,000 threshold
FTA first entry	Year FE	0.002**
		(0.001)
FTA first entry	HS2-year FE	-0.000
		(0.001)
FTA first entry	HS4-year FE	0.001
		(0.001)
First entry	Year FE	0.004***
		(0.001)
First entry	HS2-year FE	0.003*
		(0.001)
First entry	HS4-year FE	0.006***
		(0.002)
Annual presence	Year FE	0.014***
		(0.004)
Annual presence	HS2-year FE	0.010**
		(0.004)
Annual presence	HS4-year FE	0.019***
		(0.005)

Notes: The table uses a $10,000 entry threshold. The risk set contains HTS8 products with zero U.S. imports from Jordan in 1998–2000 and a clean FTA tariff schedule. FTA-first-entry equals one only when the first year in which total U.S. imports cross the threshold is also a year in which FTA imports cross the threshold. First-entry models retain product-years until first entry; annual-presence models retain the full product-year panel. Liberalization is cumulative FTA tariff liberalization divided by 10. Standard errors are clustered by HTS8. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2 shows that FTA liberalization predicts entry into the U.S. market. In the pre-

ferred first-entry specification with HS4-by-year fixed effects, a 10-percentage-point tariff concession raises the probability of first entry by 0.006 probability units, or about 0.6 percentage points, at the $10,000 threshold. The estimate remains positive when using only year fixed effects or HS2-by-year fixed effects, although the magnitude is smaller. The annual-presence outcome gives a similar pattern: larger tariff concessions are associated with a higher probability that a previously absent product is observed in U.S. imports.

Appendix Table A4 shows that the liberalization result is robust when entry is defined using the lower $5,000 threshold. The appendix also reports a $15,000 threshold as an additional sensitivity check. The broader first-entry and annual-presence outcomes remain positive across thresholds and fixed-effect specifications. The stricter FTA-first-entry outcome is weaker once sector-by-year fixed effects are included, so I interpret the entry evidence as strongest for the association between FTA tariff liberalization and product entry into the U.S. market, and more cautious on whether first observed entry occurs directly under the FTA program.

To sum up, the results indicate that relaxing non-tariff barriers to trade, such as rules-of-origin and production-location requirements, through the introduction of a new preferential trade regime such as the U.S.–Jordan FTA can deepen trade along three distinct margins. First, it induces exporters to reallocate existing trade toward the new preferential regime. Second, it expands exports to the destination market, particularly in industries that already have an established presence in that market. Third, it encourages entry at the extensive margin by enabling products that were not previously exported to the destination market to begin entering it.

6 Conclusion

Eligibility requirements embedded in preferential trade agreements can limit the gains from improved market access. This paper examines what happens when exporters gain access to a less restrictive preferential regime. I show that the introduction of the U.S.–Jordan FTA provided Jordanian exporters with a less restrictive alternative to the existing QIZ program. As the FTA was gradually implemented, exporters shifted from using QIZ toward the FTA. The FTA also expanded exports to the United States, with the largest increases occurring among products in industries with a strong pre-existing presence in the U.S. market. Suggestive evidence indicates that exports to non-U.S. destinations did not decline following liberalization, consistent with trade expansion rather than a reallocation of existing exports away from other markets. Finally, by increasing the profitability of serving the U.S. market, FTA liberalization encouraged the entry of new products into the U.S. market.

Together, these findings highlight the importance of the design of rules of origin, production-location requirements, and other eligibility conditions within preferential trade agreements. These considerations are particularly relevant for countries in the Middle East and North Africa, where intra-regional trade remains limited despite the presence of preferential trade agreements.

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A Appendix

Table A1: Comparison of QIZ and FTA Eligibility Requirements

	QIZ	**U.S.–Jordan FTA**
Qualifying-content requirement	35 percent	35 percent
Composition	At least 11.7% Jordanian content, 8% Israeli content, and the remaining 15.3% from an eligible set of sources	Jordanian materials and processing, with U.S.-origin materials permitted to account for up to 15 percentage points
Mandatory Israeli sourcing	Yes: at least 8%	None
Eligible production location	Designated QIZs only	Anywhere in Jordan

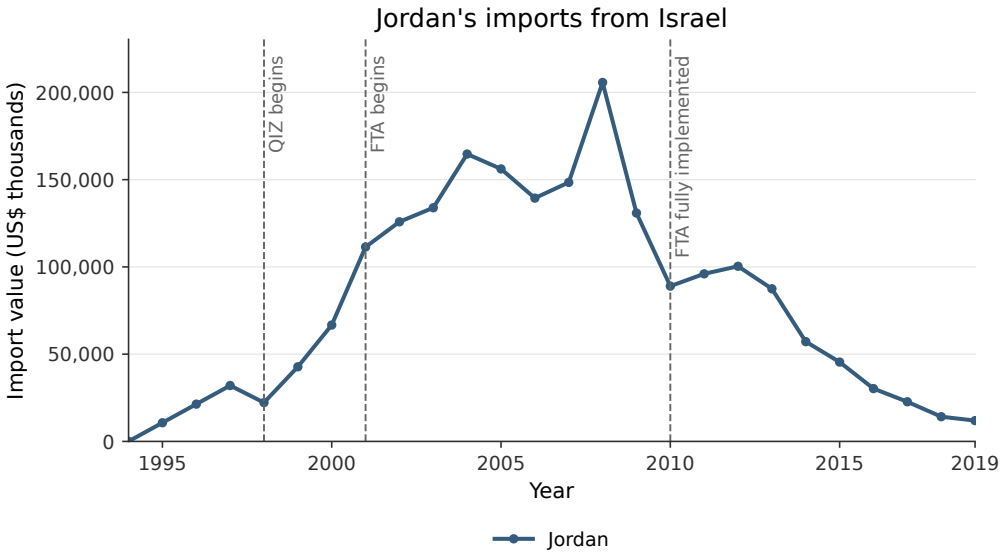


Figure A1: Jordan's Imports from Israel

Notes: The figure reports Jordan's total imports from Israel in thousands of current U.S. dollars. Vertical dashed lines mark the beginning of the QIZ program, the entry into force of the U.S.–Jordan FTA, and full implementation of the FTA tariff phaseout. The series follows the sample window used in the original synthetic-control figure and excludes the synthetic-control counterfactual.

Table A2: Export Growth Around Zero FTA Tariffs

Event time	Estimate	Std. error
-3	0.120	(0.155)
-2	-0.359*	(0.211)
-1	0.112	(0.289)
+1	0.565**	(0.243)
+2	0.529**	(0.262)
+3	0.786***	(0.174)
+4	0.654**	(0.319)
+5	0.248	(0.229)
+6	0.455***	(0.174)
+10	0.655***	(0.200)
Reference event time	0	
Fixed effects	HS6-destination, HS6-year, destination-year	
Cluster	HS6	

Notes: The table reports the interaction-weighted Sun–Abraham-style event-study estimates corresponding to Figure 6. The outcome is $\log(1 + exports_{hdt})$. Coefficients are on $\mathbf{1}\{d = US\} \times \mathbf{1}\{event = k\} \times HS2Share_s^{2000}$, where $HS2Share_s^{2000}$ is measured in 10-percentage-point units. Event time 0 is the year before the product reaches a zero FTA tariff; event time 1 is the first zero-tariff year. Standard errors are clustered by HS6. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Export-Growth Event-Study Robustness

Figures A2 and A3 re-estimate the export-growth event study using conventional dynamic TWFE and PPML specifications. Both use the same event clock as the main specification: event time 0 is the year before an HS6 product reaches a zero FTA tariff, and event time 1 is the first zero-tariff year. The figures report three pre-treatment periods and ten post-treatment periods.

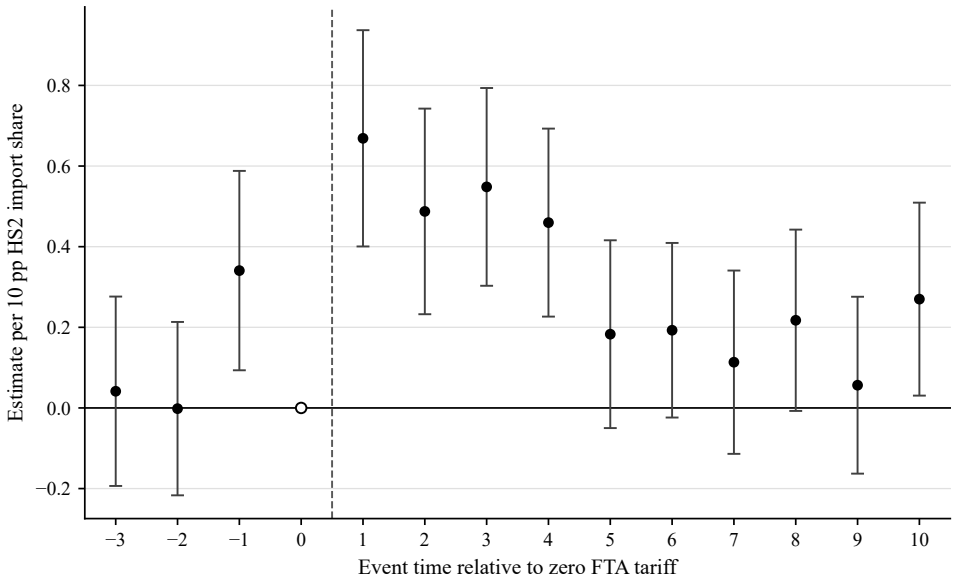


Figure A2: Export Growth Around Zero FTA Tariffs: TWFE Robustness

Notes: The figure reports coefficients from a dynamic TWFE version of equation (17). The outcome is $\log(1 + exports)$. Coefficients are measured per 10-percentage-point increase in the HS2 sector's 2000 share of Jordan's U.S. exports. The specification includes HS6-by-destination, HS6-by-year, and destination-by-year fixed effects. Standard errors are clustered by HS6 product. Bars report 95 percent confidence intervals.

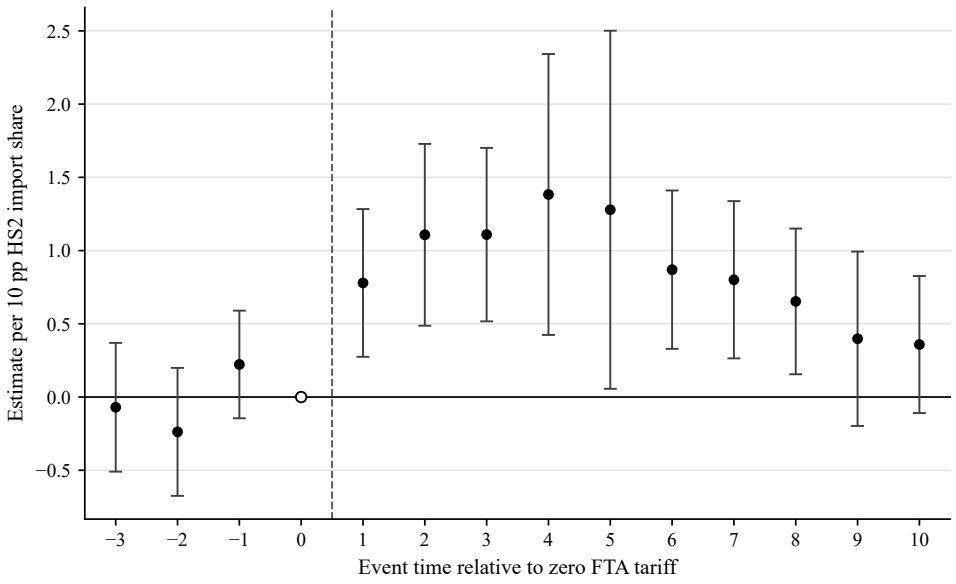


Figure A3: Export Growth Around Zero FTA Tariffs: PPML Robustness

Notes: The figure reports coefficients from a dynamic PPML version of equation (17). The outcome is export value in levels, preserving zero trade flows subject to fixed-effect separation. Coefficients are measured per 10-percentage-point increase in the HS2 sector's 2000 share of Jordan's U.S. exports. The specification includes HS6-by-destination, HS6-by-year, and destination-by-year fixed effects. Standard errors are clustered by HS6 product. Bars report 95 percent confidence intervals.

Trade Expansion versus Destination Reallocation

To distinguish export expansion from destination reallocation, I re-estimate the zero-tariff event-study design separately for three outcomes: exports to the United States, exports to non-U.S. destinations, and total exports. The unit of observation is the HS6 product-year. Treatment timing is the first year in which the corresponding FTA tariff reaches zero, with event time 0 defined as the year before zero-tariff treatment. The coefficient of interest interacts event time with the HS2 sector's share of Jordan's U.S. exports in 2000. This asks whether products in sectors with stronger pre-existing U.S. market access grow differentially after zero-tariff liberalization, and whether that growth is accompanied by falling non-U.S. exports.

The baseline allocation exercise without product-specific trends shows similar post-zero increases, but it also displays non-flat pre-treatment coefficients. For this reason, Figure A4 reports the preferred version controlling for HS6-product specific linear trends. U.S. exports rise after products reach zero FTA tariffs, and total exports also increase in the early post-treatment years. Non-U.S. exports do not systematically fall over the same horizon. I therefore interpret the results as pointing toward trade expansion rather than export reallocation across destinations.

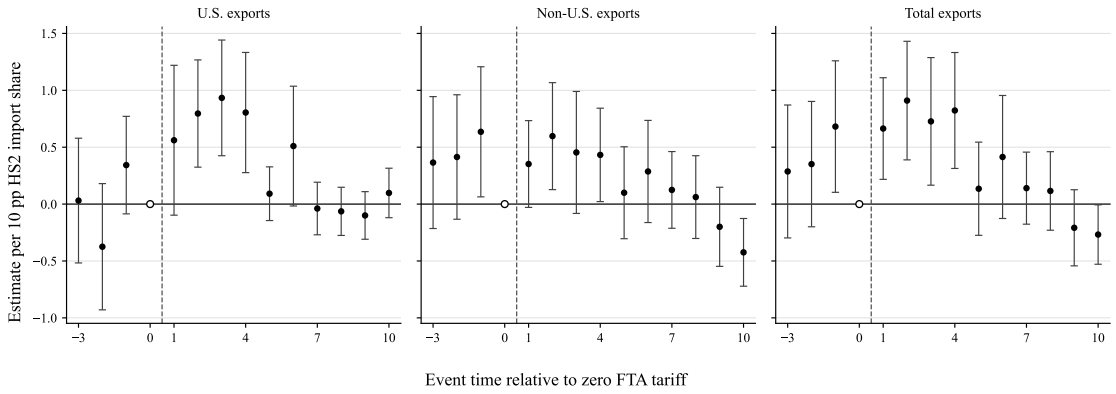


Figure A4: Export Expansion versus Destination Reallocation with HS6 Trends

Notes: The figure reports event-study coefficients for U.S. exports, non-U.S. exports, and total exports. The outcome is $\log(1 + exports)$. Coefficients are measured per 10-percentage-point increase in the HS2 sector's 2000 share of Jordan's U.S. exports. Event time 0 is the year before the product reaches a zero FTA tariff. The specification includes HS6 fixed effects, year fixed effects, and HS6-specific linear trends. Bars report 95 percent confidence intervals.

Table A3: FTA Tariff Concessions and MFN-to-FTA Take-Up: Strict Never-QIZ Robustness

	OLS share	Value-weighted share	Stacked PPML
Phaseout window, 1998–2010			
FTA liberalization (10 pp)	0.443***	0.646***	15.334**
	(0.135)	(0.238)	(7.774)
Observations	228	228	78
HTS8 clusters	114	114	13
Extended window, 1998–2022			
FTA liberalization (10 pp)	0.186	0.092	-3.127
	(0.140)	(0.198)	(2.538)
Observations	333	333	236
HTS8 clusters	114	114	26

Notes: This robustness table further restricts the ex-ante MFN risk set to 114 HTS8 products with no QIZ imports at any point during 1998–2022. This sample is cleaner mechanically but conditions on future behavior after treatment; it should therefore be interpreted as a diagnostic robustness exercise rather than the main H2 test. The dependent variable in the share specifications is $FTA_{ht}/(FTA_{ht} + MFN_{ht})$. The stacked PPML specification uses separate FTA and MFN customs values with HTS8-by-year, HTS8-by-program, and program-by-year fixed effects.

Product Entry Threshold Robustness

Appendix Table A4 re-estimates the product-entry specifications using a lower $5,000 entry threshold as the main robustness check. It also reports a $15,000 threshold as an additional sensitivity check. The table reports FTA-first-entry, first-entry, and annual-presence outcomes under year, HS2-by-year, and HS4-by-year fixed effects.

Table A4: FTA Tariff Liberalization and New U.S. Product Entry: Threshold Robustness

Outcome	Fixed effects	$5,000 threshold	$15,000 threshold
FTA first entry	Year FE	0.002***	0.002***
		(0.001)	(0.001)
FTA first entry	HS2-year FE	0.001	-0.000
		(0.001)	(0.001)
FTA first entry	HS4-year FE	0.002	0.001
		(0.001)	(0.001)
First entry	Year FE	0.003**	0.004***
		(0.001)	(0.001)
First entry	HS2-year FE	0.002	0.003**
		(0.002)	(0.001)
First entry	HS4-year FE	0.006***	0.006***
		(0.002)	(0.002)
Annual presence	Year FE	0.014***	0.015***
		(0.004)	(0.003)
Annual presence	HS2-year FE	0.010**	0.011***
		(0.004)	(0.004)
Annual presence	HS4-year FE	0.020***	0.021***
		(0.005)	(0.005)

Notes: The table reports robustness to alternative entry thresholds. The risk set contains HTS8 products with zero U.S. imports from Jordan in 1998–2000 and a clean FTA tariff schedule. FTA-first-entry equals one only when the first year in which total U.S. imports cross the stated threshold is also a year in which FTA imports cross the threshold. First-entry models retain product-years until first entry. Annual-entry models retain the full product-year panel. Liberalization is cumulative FTA tariff liberalization divided by 10. Standard errors are clustered by HTS8. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

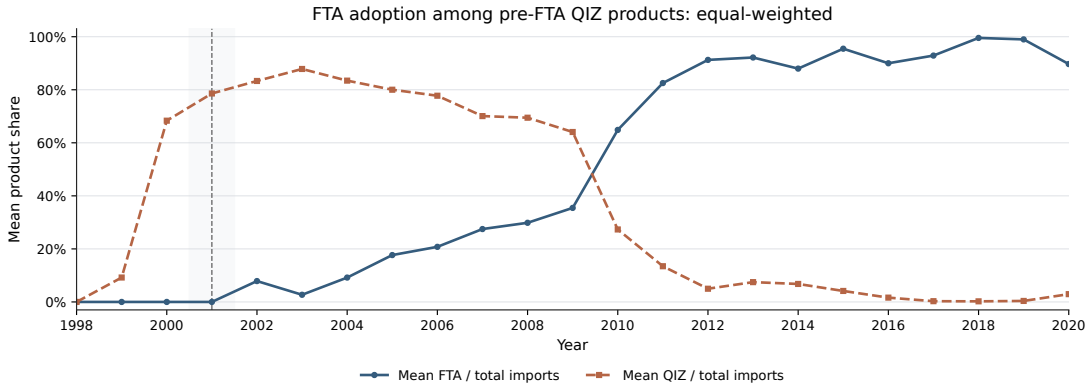


Figure A5: FTA Adoption among Pre-FTA QIZ Products: Equal-Weighted

Notes: The figure reports equal-weighted mean program shares across active HTS8 products with positive QIZ imports in 1999 or 2000. Each product receives equal weight. The shaded area marks the partial implementation year 2001.

FTA Take-Up Relative to Non-FTA Imports

Figure A6 re-estimates the broader FTA-versus-non-FTA switching exercise using the cohort-interaction event-study logic of Sun and Abraham (2021). This matters because zero-tariff timing is staggered across products, so a conventional TWFE event study may compare newly treated products with products that have already reached zero tariffs. The estimating sample and event clock are the same as in Figure 4: the sample includes all HTS8 products with a clean FTA tariff schedule, non-FTA imports combine QIZ, MFN/no-preference, and other programs, and event time 0 is the year before the product reaches a zero FTA tariff.

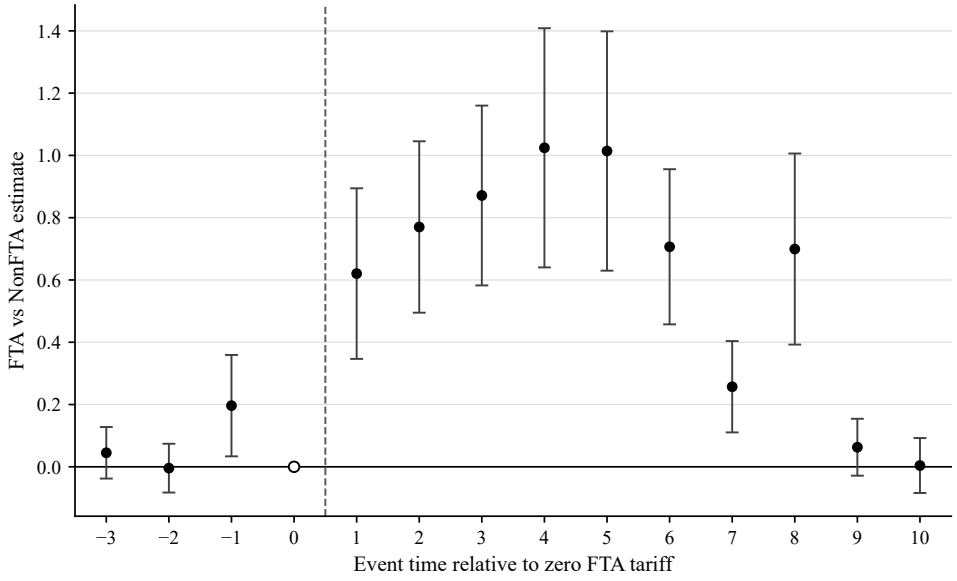


Figure A6: FTA Take-Up Relative to Non-FTA Imports: Sun–Abraham Robustness

Notes: The figure reports Sun–Abraham-style event-study estimates using HTS8-year observations. The outcome is $\log(1 + FTA) - \log(1 + NonFTA)$, where non-FTA equals QIZ plus MFN/no-preference plus other program imports. Event time 0 is the year before the product reaches a zero FTA tariff. The specification includes HTS8 fixed effects, year fixed effects, and HS6-specific linear trends. Standard errors are clustered by HTS8. Bars report 95 percent confidence intervals.